CoRoT-2b

R-0160

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_180622 · created 2026-07-16T15:30:27+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458291.7434 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.199 +/- 0.012
Transit depth (Rp/Rs)^2	3.98 +/- 0.5 [%]
Orbital Inclination (inc)	89.78 +/- 0.83
Airmass coefficient 1 (a1)	1.133 +/- 0.024
Airmass coefficient 2 (a2)	-0.0964 +/- 0.0049
Scatter in the residuals of the lightcurve fit is	2.09 %
Best Comparison Star	None
Optimal Aperture	4.6
Optimal Annulus	14.05
Transit Duration (day)	0.0995 +/- 0.0012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.199 ± 0.012 vs 0.1667 (deviation 2.22σ)
Duration fit vs published	0.0995 d vs 0.095 d (deviation 3.09σ)
Mid-time O–C	-5.9 min
Depth SNR	13.7
Residual scatter	2.09 %

Light curve

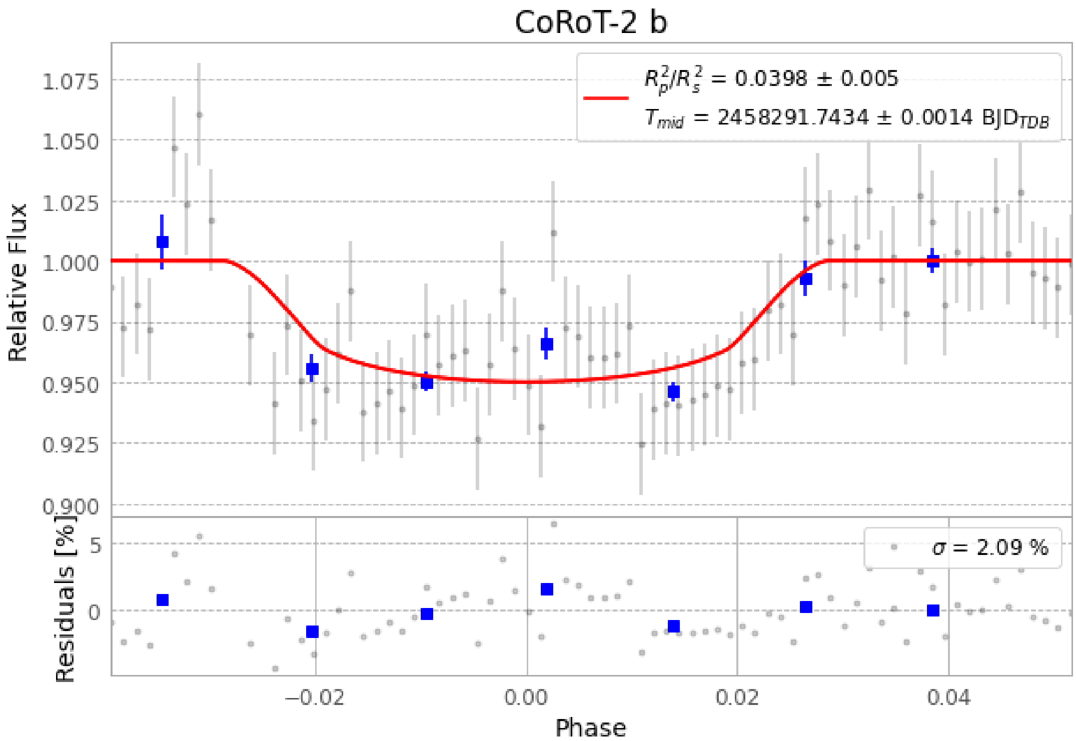

Figure 1 — FinalLightCurve_CoRoT-2 b_2018-06-21.png (SHA-256 056aff969410ed8f…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [319, 142], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 6f77ec0b8d81d023…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1634555`

Data provenance

77 FITS frames (51 MB), `CoRoT-2180622040312.FITS` → `CoRoT-2180622075112.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-180622.log (263082536709…), FinalLightCurve_CoRoT-2 b_2018-06-21.png (056aff969410…), FinalLightCurve_CoRoT-2 b_2018-06-21.csv (1719b43d45d7…)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-16 15:30Z.